

Facile migration of potassium ions in a ternary P3-type $K_{0.5}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ cathode in rechargeable potassium batteries

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ARTICLE INFO

Keywords:

Layered
First-principles calculation
Cathode
Potassium
Batteries

ABSTRACT

A ternary P3-type $K_{0.5}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ material is introduced, herein, as a promising cathode material for K ion batteries. The disadvantages associated with Mn-based layered cathode materials – structural degradation and capacity deterioration – are overcome by the partial replacement of Mn^{3+} with Fe^{3+} and Ni^{2+} , which tends to increase the average oxidation state of Mn to 3.75+. First-principles calculation predicts the sequence of redox pairs from $Mn^{3+/4+}$, $Fe^{3+/4+}$, and $Ni^{2+/3+}$ with increasing the operating voltage to 3.9 V. $K_{0.5}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ exhibits a high reversible discharge capacity ($\sim 120 \text{ mAh g}^{-1}$) and excellent structural integrity over 300 cycles (74% capacity retention) between 1.5 – 3.9 V at 50 mA g^{-1} . This outstanding performance of $K_{0.5}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ is attributed to the slight structural variation ($\sim 4.1\%$) of its P3–O3 phase transition predicted by the first principles calculation. Surprisingly, the obtained capacity reaches 76 mAh g^{-1} at a rate of 2.5 A g^{-1} , in which the facile migration of K ions is explained by the low activation energy barrier of $\sim 438 \text{ meV}$ predicted by the nudged elastic band (NEB) method.

1. Introduction

Li-ion batteries (LIBs) are considered one of the best existing technologies for energy storage with high energy conversion efficiency [1–3]. Recent fluctuation in the price of Li and shortage of Li resources have necessitated the exploration of alternative energy storage systems that can overcome the problem of cost and reserve. Li resources are limited (only 0.0017 wt%); in contrast, Na and K resources account for 2.36 wt% and 2.09 wt% of the Earth's crust, respectively [4]. At present, since there is no other system to replace LIBs in terms of battery performance and applicability, abundance of sodium and potassium resources are very attractive to use those monovalent elements as charge carriers toward battery applications; namely, sodium-ion batteries (SIBs) and potassium-ion batteries (KIBs) because of their similar chemistries regarding the conversion process from chemical to electrical energy [5–8]. Among them, KIBs have been studied such as particularly layered K_xMeO_2 ($Me = Cr, Mn, Co, Fe, Ti, Ni$) [9–16], polyanionic compounds [17–20], organic materials [21,22], and Prussian blue [6,23]. Among

them, layered Mn-based cathode materials have attracted much interest due to their higher capacity relative to those of the other compounds shown above. Kim et al. [14] reported a P3-type $K_{0.5}MnO_2$ cathode material (space group: $R\bar{3}m$) that exhibited an initial discharge capacity of 110 mAh g^{-1} in the voltage range of 1.5–3.9 V. Further, Vaalma et al. [9] introduced a P'2-type $K_{0.3}MnO_2$ material (space group: Ccm) that exhibited a high initial discharge capacity of about 130 mAh g^{-1} in the voltage range of 1.5–4.0 V. Both K_xMnO_2 ($x = 0.3$ and 0.5) materials exhibited high initial discharge capacities; however, they experienced irreversible multiple phase transitions during their cycles, which eventually resulted in poor retention after several cycles. Since the initial average oxidation states of Mn are $3.7+$ and $3.5+$ for $x = 0.3$ and 0.5 in K_xMnO_2 , respectively, the cooperative Jahn-Teller effect of Mn^{3+} in MnO_6 may not seriously affect the crystal structure due to the high oxidation states of Mn at even deeply discharged states. Probably, the step-wise multiple phase transition associated with the K^+ /vacancy ordering is urgently required to minimize the phase transition, which would lead to improved capacity retention after the cycles. Generally, in

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the Na system, the partial substitution of Ni [24–26], Co [25,26], Cu [25,26], Zn [25], Mg [26], Ti [26,27], Al [25], or Fe [25,27,28] in Na_xMnO_2 effectively suppresses such irreversible multiple phase transitions, which enables long-term high capacity retention. These synergetic effects are expected to improve the resulting electrochemical performances, provided that they can be applied to the K system.

To obtain long-term sustainable cathodes in K cells with high capacities, we designed a ternary compound using Mn, Fe, and Ni, namely $\text{K}_{0.5}[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$, which is crystallized into a P3-type layered structure with an $R3m$ space group. Generally, the incorporation of Ni or Fe increases the resulting electrical conductivity in Na-based compounds [24,28]. Another merit of utilizing Fe is that the Jahn-Teller Fe^{4+} provide the buckling capability to reduce the migration barrier, and thereby promote the diffusion of Na^+ [29]. Expectedly, this effect can facilitate the facile diffusion of K ions within the structure. In this study, we explore the fast migration of K ions in the P3-type $\text{K}_{0.5}[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$ cathode material. Using the first-principles calculations, the diffusion of K ions is predicted with an activation energy barrier of ~ 438 meV, which is lower than the value obtained in the P2-type Na layered cathode material (typically > 500 meV) [30,31], followed by successive redox reactions of $\text{Mn}^{3+/4+}$, $\text{Fe}^{3+/4+}$, and $\text{Ni}^{2+/3+}$ pairs. Electrochemical tests in the K cells revealed that the P3-type $\text{K}_{0.5}[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$ cathode exhibits a high discharge capacity of 120 mAh g^{-1} and retains 74% of its initial capacity after 300 cycles between 1.5–3.9 V at 50 mA g^{-1} . As theoretically predicted, the $\text{K}_{0.5}[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$ material proves its eligibility as a high rate electrode by exhibiting a discharge capacity of 76 mAh g^{-1} at a rate of 2.5 A g^{-1} . *Operando* synchrotron X-ray diffraction and X-ray absorption studies are employed to understand the insertion/extraction process of the K ions in the host material.

2. Experimental section

2.1. Synthesis

The P3-type $\text{K}_{0.5}[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$ cathode material was synthesized by a typical combustion method. An aqueous solution was prepared by dissolving stoichiometric amounts of KNO_3 (Aldrich), $\text{Mn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (Aldrich), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (Aldrich), $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (Aldrich), and citric acid (Aldrich), with sucrose (Aldrich) as the chelating agent, in distilled water. The details are described in our prior works [24,28]. The obtained powder precursor was calcined at 800°C for 5 h under air exposure.

2.2. Material characterization

X-ray diffraction (XRD, X'Pert, PANalytical) with Cu-K α radiation was used to analyze the crystal structure of the as-synthesized materials. Because of readiness of moisture uptake of potassiated transition metal oxides, XRD analysis was performed using a special XRD sample folder with Mylar film (Fig. S1). The obtained XRD patterns were analyzed using the FULLPROF Rietveld program [32]. The particle morphologies of the products were observed via field-emission scanning electron microscopy (FE-SEM; JXA-8100, JEOL) with energy dispersive X-ray spectroscopy (EDS; 7200-H, HORIBA) and transmission electron microscopy (TEM, JEM2010, JEOL). The chemical compositions of the synthesized materials were analyzed by inductively coupled plasma-atomic emission spectroscopy (ICP-AES, OPTIMA 8300, PerkinElmer). The potassium storage mechanisms during the charge (oxidation) and discharge (reduction) processes were investigated by *operando* synchrotron X-ray diffraction (*o*-SXRD) at the 3D beamlines of the Pohang Accelerator Laboratory (PAL), Pohang, South Korea. The data were collected on an MAR345 image plate with an incident wavelength of $1.0332 \text{ \AA}$ (12 keV) [33]. The *ex-situ* X-ray absorption near-edge structure (XANES) was carried out at the 6D beamline of the Pohang Accelerator Laboratory (PAL), Pohang, South Korea. The direct voltamperic method (CMT-SR1000, IT) was used to measure the direct current (DC) electrical

conductivity, where disc samples were in contact with a four-point probe. X-ray photoelectron spectroscopic analysis (XPS, PHI 5600, PerkinElmer) was performed to investigate the oxidation states at the surface of the $\text{K}_{0.5}[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$ in the macro mode ($3 \times 3 \text{ mm}^2$) using Mg K α and Al K α as X-ray sources to avoid overlapping of binding energies with Auger lines.

2.3. Electrochemical tests

The electrodes were prepared by blending the active materials (80 wt.%), conducting carbon (Super-P; KS-6 = 1: 1, 10 wt.%), and polyvinylidene fluoride (PVDF, 10 wt.%) in *N*-methyl-2-pyrrolidone (NMP). The loaded slurry (diameter: 14 ϕ , and mass loading: $\sim 3.0 \text{ mg cm}^{-2}$) was applied onto Al foils, and then dried at 110°C overnight in a vacuum oven. The half cells were assembled using R2032 coin-type cells with K metal as the anode (Sigma Aldrich, 99.5%), separated by a glass fiber (AdvanTec) in an Ar-filled glove box. The electrolyte solution was comprised of 0.5 M KPF $_6$ in ethylene carbonate (EC): diethyl carbonate (DEC) (1:1 volume ratio). The assembled cells were tested between 1.5–3.9 V at 25°C . After assembling the cell, the GITT measurements of the electrode were conducted at a rate of 50 mA g^{-1} with a 30-min charge (depotassiation) or discharge (potassiation) step, with 10-min rest and open-circuit periods of 1 h. The full cells were assembled by pairing with pre-potassiated hard carbon anode (Kureha) which were punched out of the dried electrodes onto Cu foil (diameter: 16 ϕ , and mass loading: $\sim 1.8 \text{ mg cm}^{-2}$). The capacity ratio of anode and cathode was balanced to ~ 1.2 , and full cells were separated by a glass fiber (AdvanTec) in a R2032 coin cell.

2.4. Computational details

Density functional theory (DFT) calculations were performed using the Vienna *Ab initio* simulation package (VASP) [34]. We utilized projector-augmented wave (PAW) pseudopotentials [35] with a plane-wave basis set as implemented in VASP. Perdew–Burke–Ernzerhof (PBE) parametrization of the generalized gradient approximation (GGA) [36] was used for the exchange-correlation functional. The GGA + U method [37] was adopted to address the localization of the *d*-orbitals in Mn, Fe, and Ni ions, with U values of 3.9, 4.0, and 6.0 eV, respectively, as determined in previous reports [38]. All calculations were performed with an energy cutoff of 500 eV until the remaining force in the system converged to less than $0.02 \text{ eV \AA}^{-1}$ per unit cell. The cluster-assisted statistical mechanics (CASM) software [39] was used for the generation of all the K/vacancy configurations for each composition, followed by full DFT calculations on a maximum of 30 configurations with the lowest electrostatic energy for each composition to obtain a convex hull plot for P3-/O3-K $_x$ [Mn $_{0.8}$ Fe $_{0.1}$ Ni $_{0.1}$]O $_2$. Nudged elastic band (NEB) calculations [40] were conducted to determine the activation barrier of K $^+$ diffusion in the P3-type $\text{K}_{0.5}[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$ structure. A unit cell made by four formula units of $\text{K}_{0.5}[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$ was used, and one K vacancy was generated to model the K $^+$ diffusion. We considered five intermediate states between the first and the final images of a single K $^+$ diffusion event. During the NEB calculation, all the structures were allowed to relax within the fixed lattice parameters.

3. Result and discussion

The elemental composition of the synthesized product was initially characterized by ICP-AES, and the crystal structure was determined by XRD coupled with Rietveld refinement. The resulting elemental proportions of K, Mn, Fe, and Ni were determined to be 0.5: 0.8: 0.1: 0.1, respectively, by ICP-AES. The synthesized $\text{K}_{0.5}[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$ material was identified through the Rietveld refinement of the XRD patterns (Fig. 1), assuming an $R3m$ space group. Since the present compound was hygroscopic, we used a Mylar film to avoid moisture absorption, which would have caused the formation of hydrated transition metal oxides

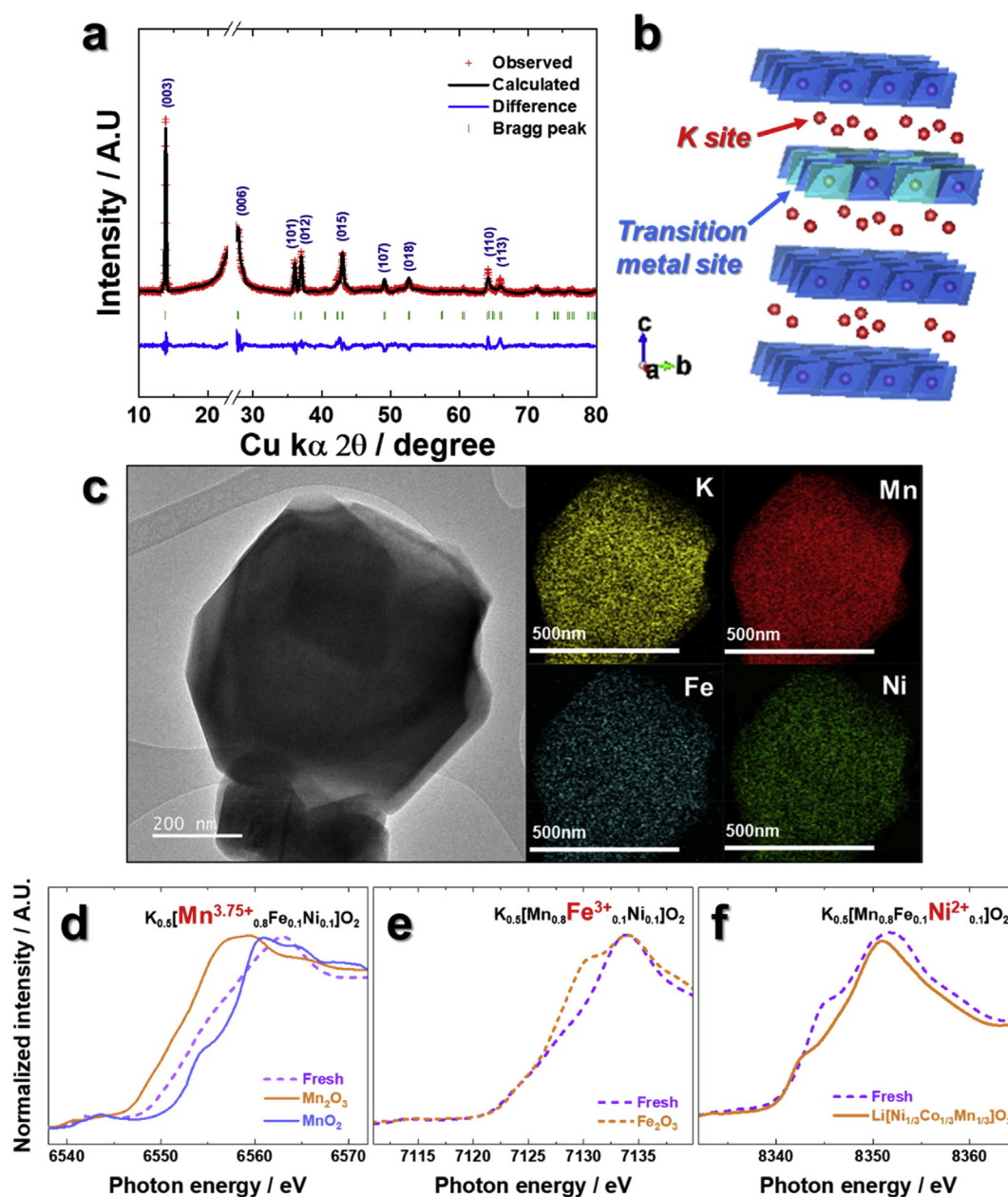


Fig. 1. (a) Rietveld refinement results for the XRD pattern; (b) schematic illustration based on the refined data; (c) TEM image with EDS mapping images for K, Mn, Fe, and Ni ions, and *ex-situ* XANES spectra for the (d) Mn K-edge, (e) Fe K-edge, and (f) Ni K-edge of the P3-type $\text{K}_{0.5}[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$ fresh powder.

(Fig. S2). A broad hill related to the Mylar film was excluded for the Rietveld refinement of the XRD patterns of the compound. The observed XRD pattern is consistent with the calculated one, showing lattice parameters of a -axis: 2.9010(3) Å and c -axis: 19.2560(6) Å. Details of the structural parameters are shown in Table S1. Based on the structural data obtained from the Rietveld refinement, the corresponding crystal structure of P3-type $\text{K}_{0.5}[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$ is depicted in Fig. 1b. P3-type $\text{K}_{0.5}[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$ crystallizes into a P3-type layered structure with trigonal prismatic sites for the accommodation of K ions in the interslab space between the transition metal layer with ABCCA oxygen packing. The morphology of the as-synthesized P3-type $\text{K}_{0.5}[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$ is observed as agglomerated hexagonal-platelets with average diameters in the range of 400–600 nm, as shown in the TEM (Fig. 1c) and SEM (Fig. S3) images. Homogeneous distributions of K, Mn, Fe, and Ni are observed in the elemental mapping images (Fig. 1c and S2). XANES analysis was used to confirm average oxidation states of transition metal elements in the P3-type $\text{K}_{0.5}[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$

compound (Fig. 1d–f), which was compared with references such as Mn_2O_3 , MnO_2 , $\text{Li}[\text{Ni}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}]\text{O}_2$, and Fe_2O_3 for linear fit. The results indicate that the oxidation state of Mn, Fe, and Ni were 3.75+ (Fig. 1d), which is a mixed state of Mn^{3+} and Mn^{4+} , 3+ (Figs. 1e), and 2+ (Fig. 1f), respectively. The tendency shown in XANES was also observed in XPS data (Fig. S4). For replacement type substitution, the ionic radii between elements should be within 15% [41]. In consideration of ionic radius of Mn^{4+} (0.53 Å) versus Ni^{2+} (0.69 Å), Fe^{3+} (0.645 Å), or Mn^{3+} (0.645 Å), the difference in ionic radius is over 15% for Mn^{4+} substitution. This means difficulty of replacement of Mn^{4+} by Ni^{2+} and Fe^{3+} ; however, due to the similarity in the ionic radius, these elements readily substitute Mn^{3+} . As a result, the total concentration of Mn decreased, which affects increase in the average oxidation state of Mn to 3.75+, compared to $\text{K}_{0.5}\text{MnO}_2$ (3.5+). This is further reflected in the lattice parameters. The calculated lattice parameters of $\text{K}_{0.5}[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$ ($a = 2.9010(3)$ Å and $c = 19.2560(6)$ Å) are larger than those of $\text{K}_{0.5}\text{MnO}_2$ ($a = 2.875$ Å and $c = 19.085$ Å) [14], as a

result of the replacement of Mn^{3+} by larger Ni^{2+} and Fe^{3+} , resulting in increase in the concentration of Mn^{4+} in the compound. Comparison of XANES spectra explains the increase in the oxidation state of Mn in $\text{K}_{0.5}[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$, compared to $\text{K}_{0.5}\text{MnO}_2$ (Fig. S5). To check occupancy of Ni and Fe in the Mn site, we further simulated $[\text{K}_{0.5}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{Mn}_{0.8}\text{O}_2$ structure that shows the Ni and Fe ion cannot be occupied in the K prismatic sites, leading to no formation of P3 type layered structure (Fig. S6).

The K intercalation in the K cells was investigated in the voltage range of 1.5–3.9 V at 50 mA g^{-1} . The first charge-discharge curve of the P3-type $\text{K}_{0.5}[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$ electrode indicated a lower first charge capacity (66 mAh g^{-1}) than the discharge capacity (120 mAh g^{-1}), with a coulombic efficiency of 181% ascribed to the K deficiency in the P3-type $\text{K}_{0.5}[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$ cathode material (Fig. 2a and b). Interestingly, the present P3-type $\text{K}_{0.5}[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$ electrode exhibited a smooth voltage step on charge and discharge compared to the cases of P3-type $\text{K}_{0.5}\text{MnO}_2$ [14], and P'2-type $\text{K}_{0.3}\text{MnO}_2$ [9], which may be related to the suppression of the step-wise multiple phase transition owing to the replacement of Mn by Fe and Ni. The abrupt voltage decays observed between 2.2–2.6 V on charge and 2.0–2.3 V on discharge are due to the K/vacancy ordering. The simplicity of the voltage profile is likely to

affect the improvement of the cycle performances; namely, exhibiting 74% retention of the initial discharge capacity after 300 cycles at 50 mA g^{-1} (Fig. 2a and b). The rate performances of the P3-type $\text{K}_{0.5}[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$ cathode material is acceptable in K cells, indicating that the delivered discharge capacity was 76 mAh g^{-1} even at 2.5 A g^{-1} (Fig. 2c and d). Surprisingly, the continuous cycling performed at 2.5 A g^{-1} presents a retention of $\sim 72\%$ after 200 cycles. This outstanding performance of the P3-type $\text{K}_{0.5}[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$ electrode may be associated with its high electrical conductivity, $1.5 \times 10^{-5} \text{ S cm}^{-1}$ (Table S1), and suppressed irreversible multiple phase transition observed from the charge-discharge profiles.

Operando synchrotron XRD (O-SXRD) analysis was carried out to investigate the structural evolution of P3-type $\text{K}_{0.5}[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$ during the K^+ de/intercalation processes (Fig. 3a). During charging, the P3 (003) and P3 (006) peaks monotonically shifted toward lower angles; however, P3 (101), P3 (012), P3 (110), and P3 (113) peaks gradually shifted toward higher angles in the XRD patterns. From 3.5 V to 3.9 V ranging from $\text{K}_{0.34}[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$ to $\text{K}_{0.25}[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$ (Fig. 3b), a new peak belonging to a (104) peak of O3 phase, appeared. Conversely, the P3 (105) peak was not visible in the range, indicating a phase transition from P3 to the O3 phase. The O3 phase was maintained

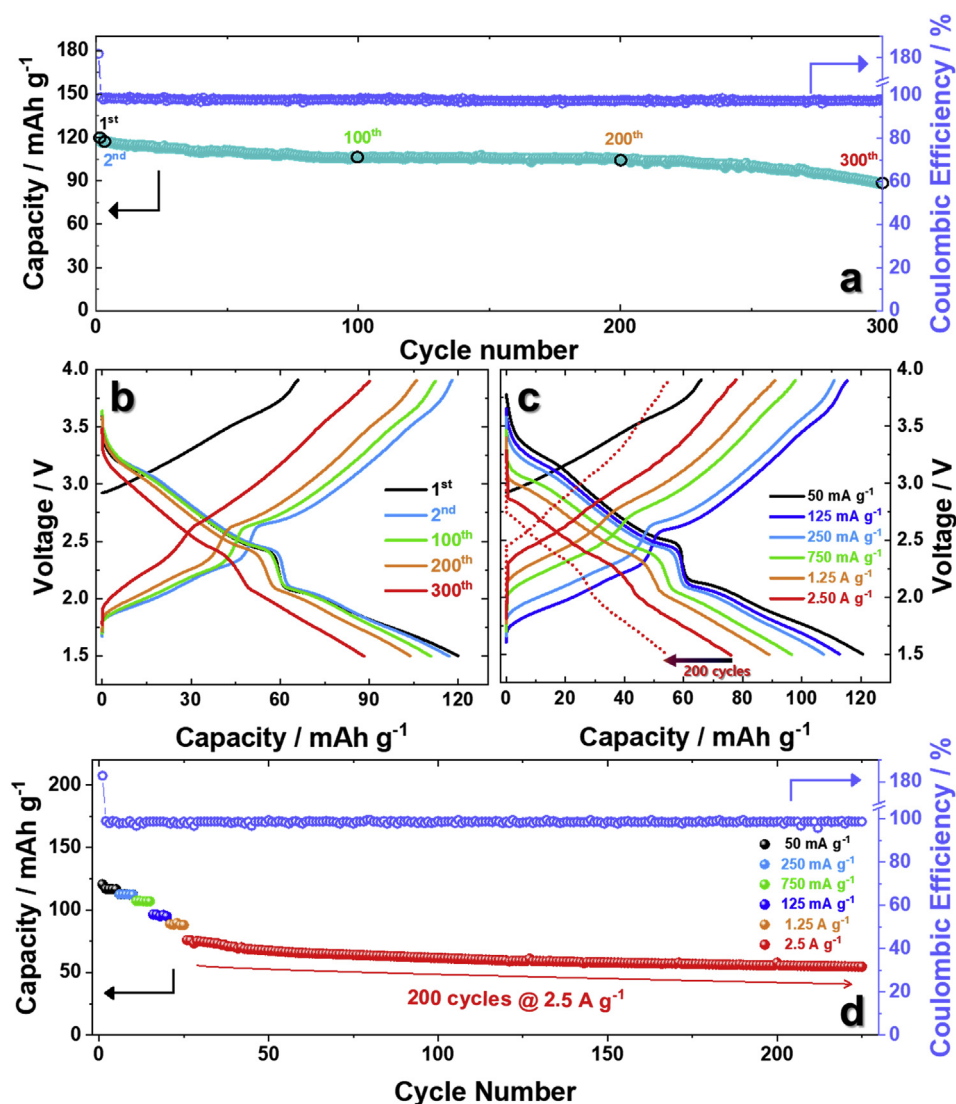


Fig. 2. (a) Cycle performances with coulombic efficiency at 50 mA g^{-1} during 300 cycles; (b) charge-discharge profiles at 50 mA g^{-1} ; (c) charge-discharge profiles at various current rates (50 mA g^{-1} , 125 mA g^{-1} , 250 mA g^{-1} , 750 mA g^{-1} , 1.25 A g^{-1} , and 2.5 A g^{-1}); and (d) the resulting rate capability of the P3-type $\text{K}_{0.5}[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$ electrode.

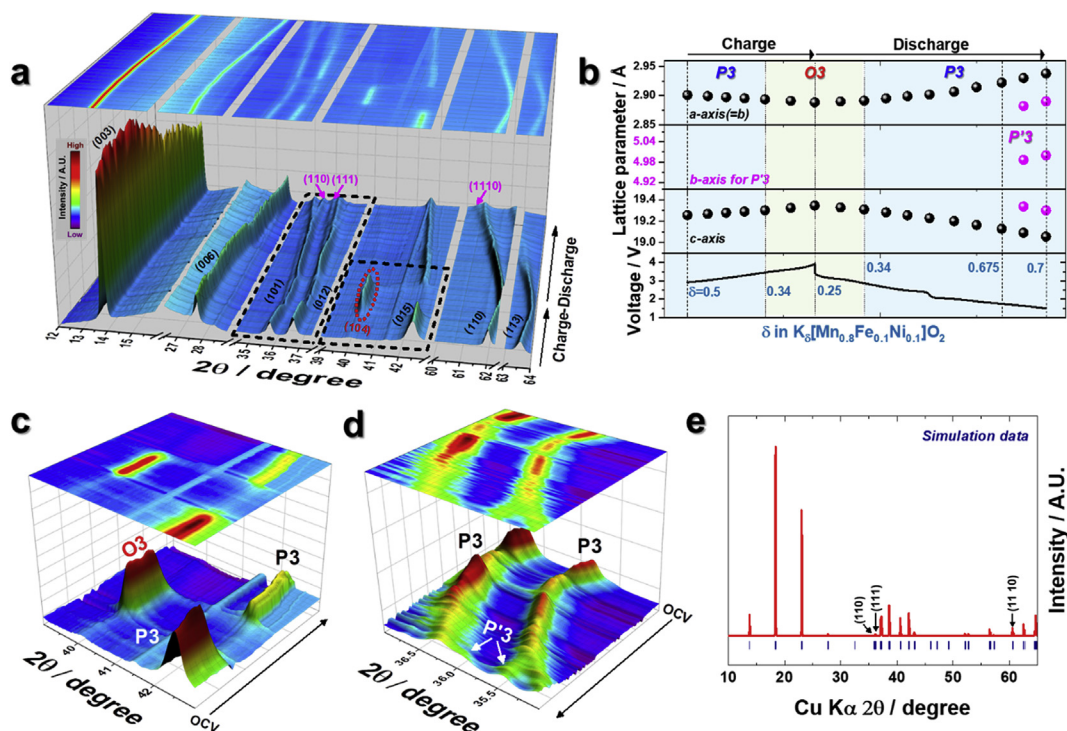


Fig. 3. (a) Operando SXRD patterns for the P3-type $K_{0.5}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ electrode with Kapton film: 17° – 30° ; (b) the lattice parameters calculated from operando SXRD patterns from $K_{0.25}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ to $K_{0.7}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ during the charge-discharge processes. Magnified operando SXRD patterns in the (c) range of 39° – 43° and (d) range of 34.5° – 37° , respectively. (e) XRD pattern of the P'3-type $K_{0.7}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ in a simulated environment.

until the charging process to 3.9 V was completed. According to earlier work on $K_{0.5}MnO_2$ by Kim et al. [14] an unknown phase presented as the “X” phase, similar to the O3 phase stacking, was dominant in the range of K_xMnO_2 ($x = 0.27$ – 0.34) on charge, while such a new phase was not observed for the present $K_x[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ ($x = 0.25$ – 0.34). This indicates that such additional phase transition from O3 to X phase was suppressed due to the substitution of Mn^{3+} site by Fe and Ni in $K_{0.5}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$.

On discharge, the newly formed O3 phase was transformed to the original P3 structure, when the K content exceeded $x = 0.34$ in $K_x[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$. Moreover, the P3 phase was maintained to $x = 0.675$ in $K_x[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$, while a gradual shift of the diffraction peaks was evident in this range. Notably, for $x > 0.675$ in $K_x[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$, peak splitting was evident (Fig. 3c), which is owing to the distortion of the P3 phase; namely, the P'3 phase with a $C2/m$ space group that is more visualized and simulated in Fig. 3d and e, respectively. Similar phenomena were observed in the P2- and P'2-type $Na_{2/3}MnO_2$ compounds that were transformed to P'2 and P''2, respectively, after deep sodiation [25,26,28]. Subsequently, we observe similar results: that the P3 phase can be distorted to the P'3 phase when the intercalated content of K exceeds $x = 0.675$ in $K_x[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$. Summarizing the above results, the present P3– $K_{0.5}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ underwent step-wise phase transitions such as P3 → O3 on charge and O3 → P3 → P'3 on discharge, of which each phase was dominant in specific ranges; in detail, P3 phase for $K_x[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ ($x = 0.34$ – 0.5), O3 phase for $K_x[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ ($x = 0.25$ – 0.34), and P'3 phase for $K_{0.5}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ ($x = 0.675$ – 0.7).

Through the first principles calculation, we confirmed the theoretical properties and reaction mechanism of P3– $K_x[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ for the K^+ de/intercalation. Fig. 4a shows the convex-hull plot of the formation energies of various P3- and O3– $K_x[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ compositions as functions of the K contents. When the K content in the structure was lowered from $x = 0.5$ to 0.25 , the corresponding P3– $K_{0.5}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ was more unstable than O3– $K_{0.25}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ was. This suggests that the occurrence of the P3–O3 phase transition at the

$K_{0.25}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ composition is reasonable and is consistent with the result of the operando SXRD data. In addition, it was verified that there are several stable intermittent phases between $K_0[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ and $K_1[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$, which imply a single-phase reaction with a sloppy charge/discharge curve for P3– $K_x[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$. Based on the difference in the formation energies between each composition, we calculated the theoretical redox potential range of $K_x[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ using the following equation:

$$V = - \frac{E[K_b(Mn, Fe, Ni)O_2] - E[K_a(Mn, Fe, Ni)O_2] - (x_2 - x_1)E[K]}{(b - a)F}, \quad (1)$$

where V is the average redox potential between $K_b[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ and $K_a[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$; $E[K_a[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2]$, the formation energy on the most stable configuration of $K_a[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$; $E(K)$, the formation energy of the K metal; and F , the Faraday constant. It was predicted that at a voltage range between 1.5 and 3.9 V (vs. K^+/K), the total 0.5 mol K^+ per formula unit can be de/intercalated from/into the $K_x[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$, corresponding to the specific capacity of ~ 120 mAh g^{-1} . Fig. 4b shows that the experimentally measured charge/discharge curve of P3– $K_x[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ is consistent with its predicted redox potentials in the range of K contents, $0.25 \leq x \leq 0.75$. In addition, we compared the integrated spin moments of Ni, Fe, and Mn between P3– $K_{0.75}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ and O3– $K_{0.25}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ using the first-principles calculation (Fig. 4c) to confirm the redox reaction of each transition metal in $K_x[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ during the K^+ de/intercalation. The absolute values of the total electron spin counts on Mn, Fe, and Ni ions in P3– $K_{0.75}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ were 5, 4, and 2, respectively, and they decreased after 0.5 mol K^+ deintercalation from the structure, which implies that the Mn^{3+}/Mn^{4+} , Fe^{3+}/Fe^{4+} , and Ni^{2+}/Ni^{3+} redox reactions may have occurred between P3– $K_{0.75}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ and O3– $K_{0.25}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$. These results indicate that the transition metal ions in P3– $K_{0.75}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ can provide sufficient electrons for the 0.5 mol K^+ deintercalation from the structure.

Furthermore, through the first principles calculation data, we

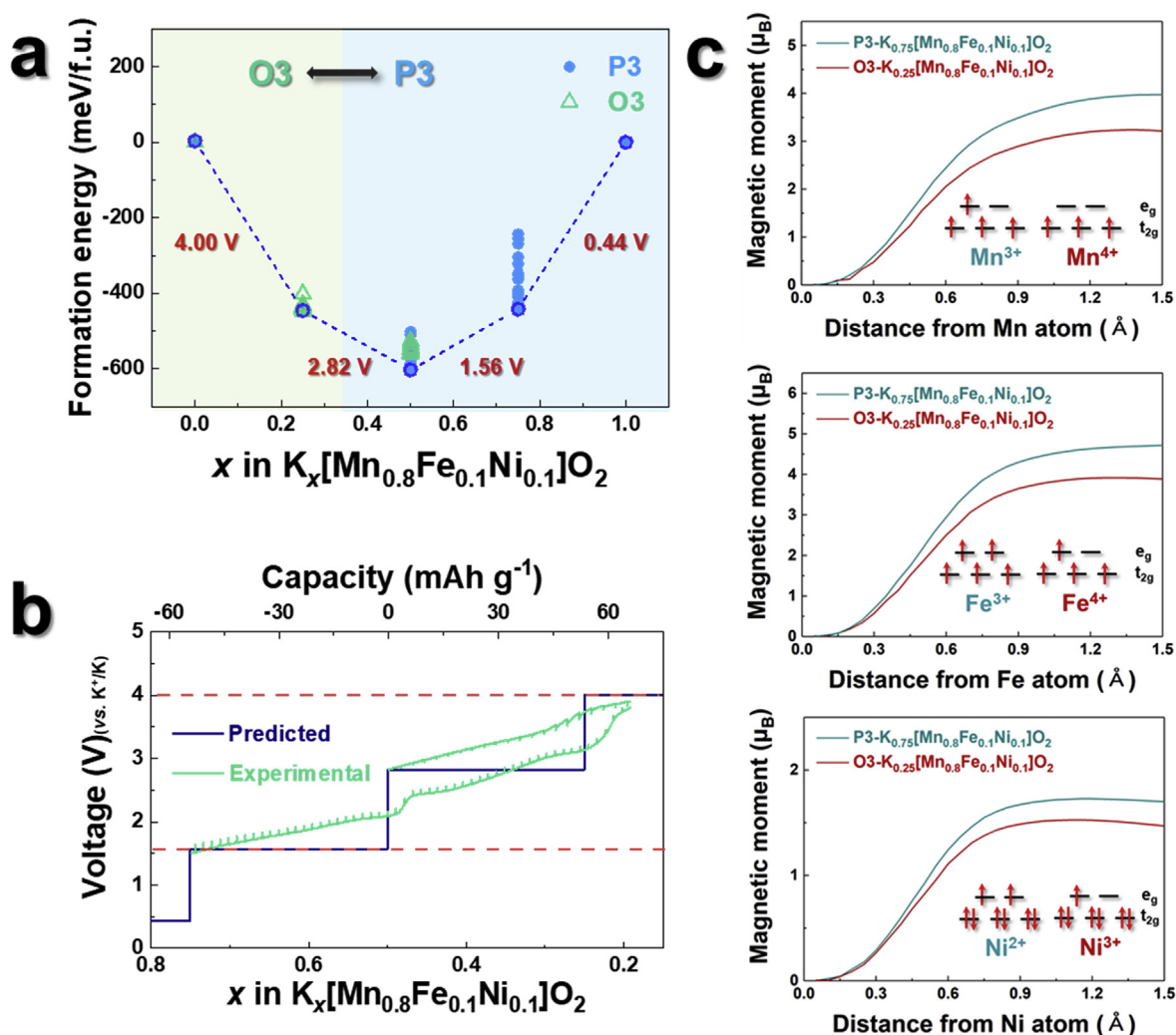


Fig. 4. (a) Convex hull plot for the formation energies of various P3-/O3- $K_x[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ electrodes, and (b) comparison of the experimentally measured GITT charge-discharge curves and the predicted voltage curves obtained using the first-principles calculations. (c) Spin integrations of Mn, Fe, and Ni ions in P3- $K_{0.75}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ and O3- $K_{0.25}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$.

predicted the structural change of P3- $K_x[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ containing K in its structure. As shown in Fig. 5a, with the deintercalation of the K^+ from P3- $K_{0.75}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$, the c -lattice parameter gradually increased from ~ 18.89 to ~ 19.10 Å owing to the repulsion between the O^{2-} ions along the c -axis, as is conventional in layered-type cathode materials [15,42]. Interestingly, although P3- $K_x[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ experienced the P3–O3 phase transition of the K contents between 0.25 mol and 0.5 mol, the c -lattice parameter tended to increase slightly as the existing K contents in the structure decreased, and the predicted difference between the c -lattice parameters of P3- $K_{0.75}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ and O3- $K_{0.25}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ was calculated to be $\sim 4.1\%$. Notwithstanding the P3–O3 phase transition, it is thought that the $K_x[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ framework experienced a slight structural variation, which may have resulted in the stable performance after several cycles, as evidenced in Fig. 2. As shown in Fig. 5b, we compared the predicted structural change of P3- $K_x[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ with the experimental results obtained by *operando* SXRD with Rietveld refinement, which clearly indicates that the experimentally measured c -lattice parameters of P3-/O3- $K_x[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ were in good agreement with the first principles calculation results.

The outstanding power-capability of P3-/O3- $K_x[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ was confirmed through the first principles calculation with the nudged elastic band (NEB) method [13,15,42]. Fig. 6a presents the K^+ diffusion

pathways in P3- $K_{0.75}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ along the ab plane, and it was predicted that an activation energy barrier of ~ 438 meV is required for the K^+ diffusion within the structure. Moreover, after the P3–O3 transition, the activation energy barrier for the K^+ diffusion in the O3- $K_{0.25}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ structure was lowered to ~ 409 meV (Fig. 6b). These results from the NEB calculation suggest that, despite the large ionic size of the K^+ , facile K^+ diffusion is possible not only into the P3-phase but also the O3-phase; thus, P3- $K_x[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ can deliver the required high power, which presents it as a promising cathode material for KIBs. We calculated diffusion coefficients in range of $0.25 \leq x \leq 0.75$ in $K_x[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$, which ranged from $\sim 10^{-9} \text{ cm}^2 \text{ s}^{-1}$ for the P3 phase to $\sim 10^{-10} \text{ cm}^2 \text{ s}^{-1}$ for the O3 phase. Based on the diffusion coefficients, activation energy for K^+ diffusion was also calculated to be 16.5 kJ mol^{-1} for the P3 phase and 21.1 kJ mol^{-1} for the O3 phase (Fig. S7). This indicates slow diffusion in the O3 phase due to the reduced interlayer distance as a result of extraction of K^+ from the host structure.

In addition, through the first principles calculation, we compared the Mn–O bond distances in an MnO_6 octahedral between pristine P3- $K_{0.75}MnO_2$ and P3- $K_{0.75}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ (Fig. S8), which indicated that the MnO_6 octahedral neighboring the FeO_6 and NiO_6 octahedra in P3- $K_{0.75}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ may experience smaller anisotropic structural distortion than that neighboring other MnO_6 octahedra in

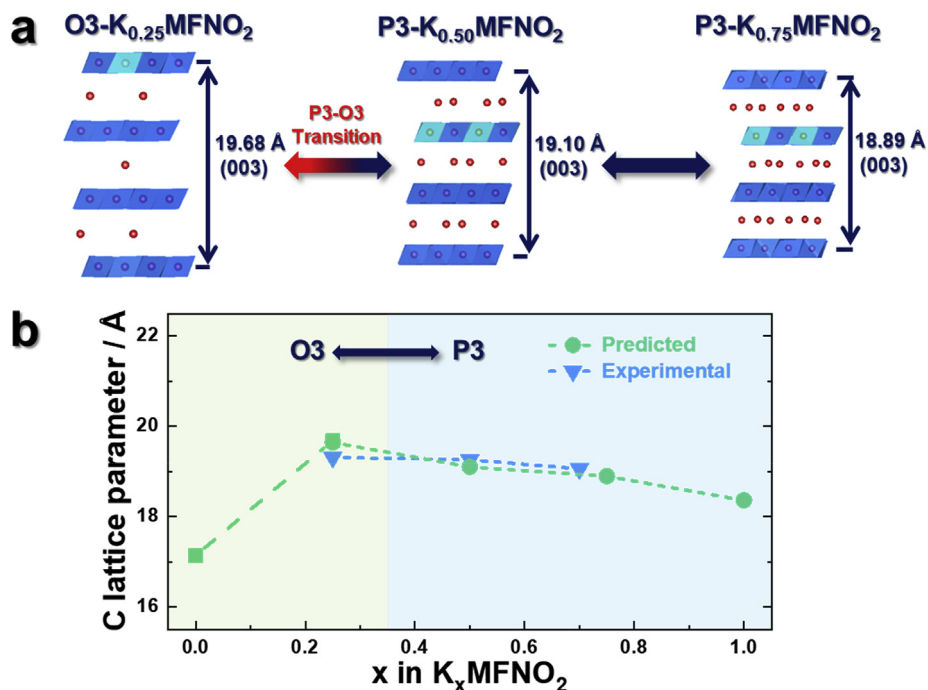


Fig. 5. (a) Predicted structural change of P3-/O3- $K_x[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$ depending on the K content in the structure, and (b) comparison of the c-lattice parameters of P3-/O3- $K_x[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$ predicted based on the first-principles calculations and experimental results.

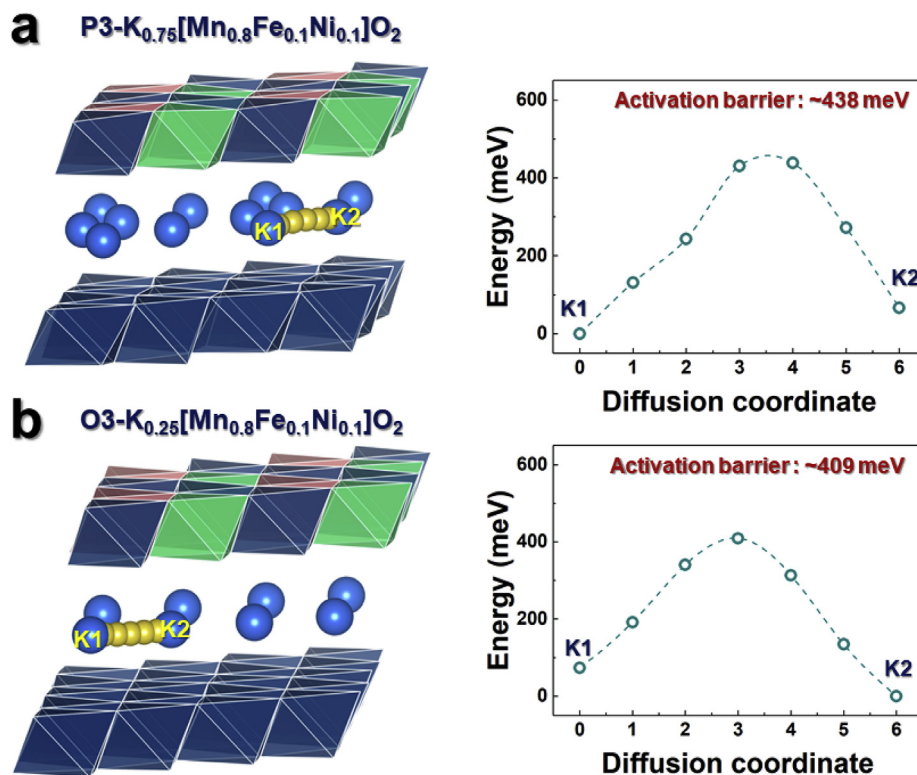


Fig. 6. K⁺ diffusion pathways and activation barriers of (a) P3-K_{0.75}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O₂ and (b) O3-K_{0.25}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O₂ obtained using NEB calculations.

pristine P3-K_{0.75}MnO₂. The existence of Fe and Ni ions in the structure can cause the suppression of the irreversible multiple phase transition.

Ex-situ XANES analysis was conducted to investigate the evolution of the average oxidation states for Mn, Fe, and Ni on samples for $x = 0.5$, $x = 0.25$ (charged), and $x = 0.7$ (discharged) in $K_x[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$.

For Mn (Fig. 7a), the oxidation occurred towards Mn⁴⁺ on charge, while the oxidation state of Mn was lowered close to Mn^{3.5+}. The value is still higher than that of Mn³⁺₂O₃. The change in the oxidation state of Mn approximated to 3.5 + is beneficial to prevent the Jahn-Teller effect from appearing in the MnO₆ octahedra. Furthermore, the MnO₆ octahedra

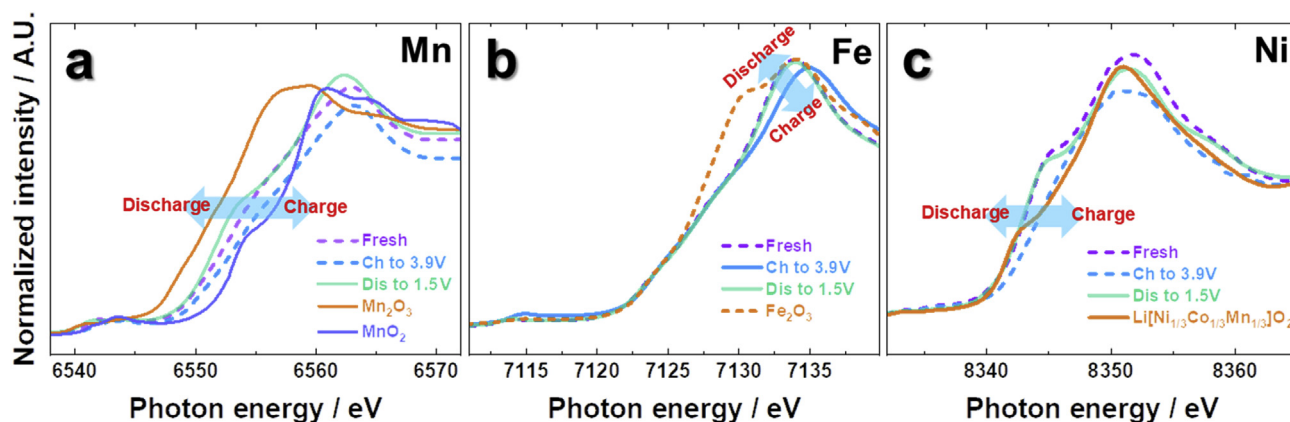


Fig. 7. *Ex-situ* XANES spectra at the (a) Mn k-edge, (b) Fe k-edge, and (c) Ni k-edge on samples with $x = 0.5$, $x = 0.25$ (charged), and $x = 0.7$ (discharged) in $K_x[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$.

share oxygen with Fe and Ni (O–Mn–O–Fe–O–Ni–O), so that such structural defect can be further suppressed by the substituents, as predicted by the first-principle calculations shown in Fig. S8. The Fe K-edge spectrum of the fresh $K_{0.5}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ showed a similar profile to that of Fe_2O_3 , indicating the oxidation state of Fe as $3+$ (Fig. 7b). By comparing the spectrum with those of $K_{0.25}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ (charged) and $K_{0.7}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ (discharged), the Fe K-edge spectrum of $K_{0.25}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ shifted towards the higher energy region compared to that of $K_{0.5}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$, indicating that Fe^{3+} is oxidized to Fe^{4+} . After discharge ($K_{0.7}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$), the Fe K-edge spectrum shifted back to the original position as a result of the reduction to Fe^{3+} . This indicates that the Fe redox pair is reversible. The reversibility of the redox reaction was also perceived for the Ni K-edge (Fig. 7c), indicating the motion of the $Ni^{2+/3+}$ redox pair during charge and discharge.

The electrode stability was further investigated by *ex-situ* XRD and TEM using a post-cycled electrode (Fig. 8). The recovered K metal anode showed that the silver color of K metal was changed to a tone of purple, and the used separator was colored to light brown (Fig. S9). These changes would be ascribed to the reaction between carbonate-based electrolyte and K metal, affect the Coulombic efficiency lower than 100%. According to recent work by Nazar et al. [43], there are the high reactivity such as unwanted side reaction between K metal and conventional carbonate solvent such as propylene carbonate (PC) or ethylene carbonate (EC)/diethyl carbonate (DEC). However, the electrode retained its high crystallinity without the formation of impurities, such as Mn–O, Fe–O, and Ni–O compounds after 300 cycles. In addition, the variation in the lattice parameters are negligible for both a - and c -axis,

showing $\Delta a = 0.005 \text{ \AA}$ and $\Delta c = 0.0136 \text{ \AA}$, compared to the case with the fresh materials. Note that the post cycled $K_{0.5}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ retained its structural integrity, thereby enabling a charge delivery of $\sim 88 \text{ mAh g}^{-1}$ at the 300th cycle. (Fig. 8a and b). This finding indicates that the irreversible multiple phase transition during the cycles was suppressed by the introduction of Fe and Ni in $K_{0.5}MnO_2$, which thus led to improved electrochemical properties. To further confirm the structural stability, the post-cycled electrodes were also investigated by TEM. The post-cycled $K_{0.5}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ electrode exhibited the hexagonal-platelets morphology without morphological degradation after 300 cycles. (inset of Fig. 8). Additionally, we fabricated and performed a $K_{0.5}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ full cell paired with hard carbon anode in the voltage range of 0.5 and 3.6 V as shown in (Fig. 9). The $K_{0.5}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ /hard carbon full cell exhibited a discharge capacity of $\sim 113 \text{ mAh g}^{-1}$ with 89% retention after 150 cycles. Notably, this finding indicates the great potential of the P3-type $K_{0.5}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ cathode material to be applied in advanced K ion batteries.

4. Conclusion

In summary, we have successfully synthesized a ternary P3-type $K_{0.5}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ material by the combustion method in an attempt to overcome the irreversible multiple phase transition due to the Jahn-teller distortion of Mn. The $K_x[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ exhibited the structural stability, with slight structural variation ($\sim 4.1\%$), predicted by the first principle calculations between $O3$ - $K_{0.25}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ and $P3$ - $K_{0.7}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$. The partial substitution of Mn with Fe and

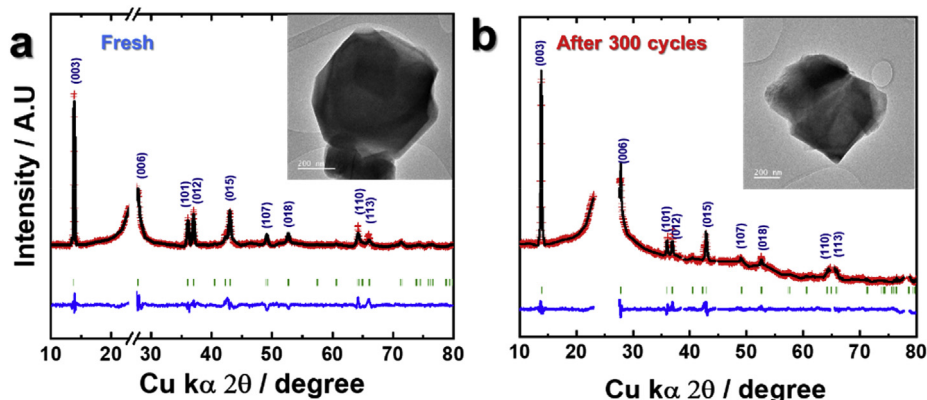


Fig. 8. Rietveld refinement patterns with TEM images for the (a) $K_{0.5}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ fresh powder and (b) $K_{0.5}[Mn_{0.8}Fe_{0.1}Ni_{0.1}]O_2$ electrode after 300 cycles.

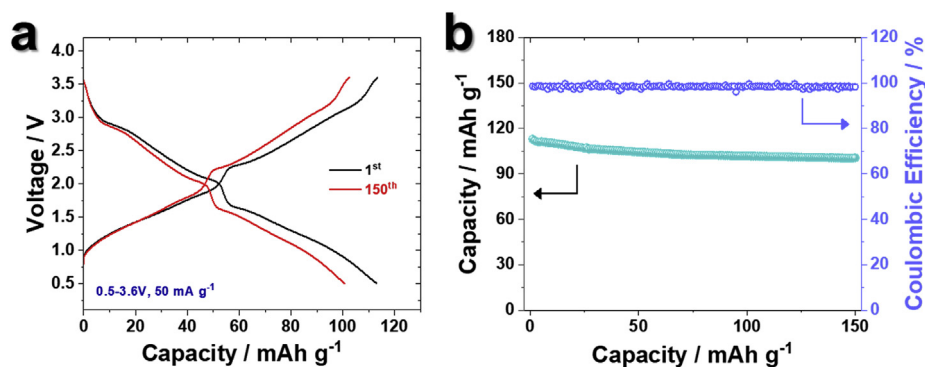


Fig. 9. Electrochemical properties of full cell paired with hard carbon anode (a) charge-discharge profile and (b) cycling performance.

Ni in $\text{K}_{0.5}[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$ facilitated the delivery of high discharge capacity ($\sim 120 \text{ mAh g}^{-1}$) with 74% capacity retention after 300 cycles and excellent rate capacities (76 mAh g^{-1}) at 2.5 A g^{-1} . This result is attributed to the low activation energy barrier required by the P3-type $\text{K}_{0.5}[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$ material for facile K^+ diffusion along the K ion diffusion pathway in the *ab* plane. Based on these results, we suggest that P3-type $\text{K}_{0.5}[\text{Mn}_{0.8}\text{Fe}_{0.1}\text{Ni}_{0.1}]\text{O}_2$ shows great promise as a cathode material for advanced KIBs.

Acknowledgments

J.U.C. and J.K. contributed equally to this work. This research was supported by the Basic Science Research Program through the National Research Foundation funded by the Ministry of Science and ICT of Korea (NRF-2015M3D1A1069713 and NRF-2017R1A2A2A05069634).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ensm.2019.09.015>.

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